



INSTRUCTION DIVISION
FIRST SEMESTER 2026-27
Course Handout Part II

Date: 02/08/2026

In addition to Part-I (General Handout for all courses appended to the timetable) this portion gives further specific details regarding the course.

Course No. : MPBA G530
Course Title : Financial Analytics
Instructor-in-charge : PABITRA BISWAS

1. Brief Course Description:

The course introduces students to the applications of different techniques of Statistics, Analytics, and machine learning to solve financial problems. The students will be exposed to understanding the characteristics of financial data, solving univariate and multivariate time series based forecasting problems for prices and returns, problems on portfolio allocation, rebalancing and optimization, Security performance analytics, trading strategies, Application of clustering and text mining in financial problem solving. The course will be a practical hands-on course which would be delivered in excel/Python.

2. Scope and Objectives of the Course:

This course aims at discussing the role of data and analytics in financial decision making, understanding the statistical properties of financial data. The student should be able to price different securities by understanding their characteristics and using different statistical and machine learning techniques. The student should also be able to create optimal portfolios catering to the requirements of different user groups and evaluate the portfolios using different metrics. The course also aims to discuss some of the most prominent applications of Analytics in Finance like Credit Scoring, Debt collection, Sentiment Analysis, Trading strategies

3. Text book: Financial Analytics (Pitbas Mohanty), Publisher- Wiley

4. Reference books:

- A. Investments (Bodie, Kane and Marcus)
- B. Quantitative Trading with R: Understanding Mathematical and Computational Tools from a Quant's Perspective (Harry Georgakopoulos)
- C. Financial Analytics with R: Building a Laptop Laboratory for Data Science by Mark J. Bennett and Dirk L. Hugen
- D. Practical Time-Series Analysis: Master Time Series Data Processing, Visualization, and Modelling using Python by Dr. PKS Prakash Dr Avishek Pal



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5. Course Plan:

Class No.	Topic	Description	(Chapter number)
1	Introduction to Financial Analytics	Introduction to Financial Analytics, Different Applications of Financial Analytics	Chapter 1, Financial Analytics
2-3	Introduction to Financial Analytics	Overview of sources of Financial Data	Chapter 2, Financial Analytics; Class notes
		Overview of Bonds, Stocks, Securities Data Cleansing	
4-6	Financial Statistics	Statistical aspects of Financial Time Series Data Plots of Financial Data (Visualizations)	Chapter3, Financial Analytics, Class notes
		Sample Mean, Standard Deviation, and Variance	
		Sample Skewness and Kurtosis, Sample Covariance	
		and Correlation, Understanding distributions of Financial Data	
7-8	Pricing and Forecasting	Univariate time series analysis for pricing and return forecasting.	Chapter 12 , Financial Analytics, Class notes
9-10	Pricing and Forecasting	Applications of Machine Learning Models for Forecasting of Price and returns	Class Notes
11-22	Portfolio Analytics	Markowitz Mean Variance optimization, CAPM- CML vs SML, time value of money	Chapter 10, Financial Analytics, Class notes
		Value at Risk (concept, excel, python)- Historical/ Monte Carlo; expected shortfall	
		Sharpe Ratio and other risk Matric	
		Portfolio Allocation Strategy and Dynamic asset allocation'	
		Portfolio Tracking	
		Portfolio Rebalancing	
		Portfolio Optimization of Two risky asset (excel/ Python)	
		Portfolio Optimization of 5 risky asset (excel/ Python)	
23-24	Clustering	Evaluating the performance of different portfolios using Risk and Return metrics	Class Notes
		Application of Clustering in Stock selection (excel/ Python)	
25-30	Credit Scoring & Debt Collection; Policy Renewal	Application of Statistical and Machine Learning models for credit scoring & Debt collection	Class Notes
31-32	Text Mining	Application of Text Mining and Natural Language processing in Financial Decision Making (impact of news headline on stock	Chapter 7 Financial Analytics, Class notes



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		price)	
33-34	Trading Strategies	Different Trading strategies using Technical Indicators, Statistical and ML models along with Back testing (Chart, excel, python)- 5 trading strategies	Chapter 11, Financial Analytics, Class Notes
35-37	Algo trading	Mechanics of Algo trading, Taxonomy of data used in Algo trading. Example of Algo with 3 strategy (excel/ python)	Class Notes
38-40	Algo trading	High Frequency Data, Working with High frequency data making 3 strategies with HF data and Back testing with past 3 months data.	Chapter 8, Quantitative Trading with R_ Understanding Mathematical and Computational Tools from a Quant's Perspective; Class Notes



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6. Evaluation Component

S. No.	Evaluation Component	Duration	Weightage (%)	Date & Time	Remarks
1	Mid-Semester Examination		30	06-Oct/ AN2	Closed Book
2	Experiential Learning Assignments / Project (2)		40		Open Book
3	Comprehensive Examination		30	7- Dec / AN	Closed Book

7. Chamber Consultation Hour: 5-6pm

8. Notice: All notices will be displayed through **Google Meet**.

Instructor-In-Charge



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